

Example 1: Brand Preference

Claim: 60% prefer Brand A. In $n = 200$, $x = 132$ prefer Brand

A ($\hat{p} = 0.66$). Test $H_0: p = 0.60$ vs $H_1: p > 0.60$ at $\alpha = 0.05$.

- Assumptions: $n \cdot p_0 = 120$, $n \cdot (1 - p_0) = 80 \geq 5$.
- $z = (0.66 - 0.60) / \sqrt{(0.60 \cdot 0.40 / 200)} \approx 2.19$.
- Critical $z = 1.645$ (right-tailed). Since $2.19 > 1.645$,

reject H_0 .

- p-value ≈ 0.014 .
- 95% CI: $0.66 \pm 1.960 \cdot \sqrt{(0.66 \cdot 0.34 / 200)} \Rightarrow (0.59, 0.73)$.
- Effect size $h \approx 0.13$ (small).

Example 2: Defect Rate Drop

$H_0: p = 0.03$ vs $H_1: p < 0.03$. Sample $n = 400$, $x = 18$ defects

($\hat{p} = 0.045$). $\alpha = 0.01$.

- $z = (0.045 - 0.03) / \sqrt{(0.03 \cdot 0.97 / 400)} \approx 2.22$.
- Because $\hat{p} > p_0$, evidence points opposite direction; fail

to reject H_0 (left-tailed test needs negative z).

- 99% CI: $0.045 \pm 2.576 \cdot \sqrt{(0.045 \cdot 0.955 / 400)} \Rightarrow (0.028, 0.062)$.
- Conclusion: No evidence for reduction; defects may be

higher than 3%.